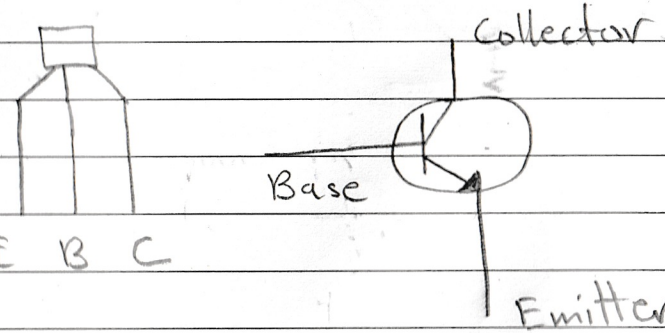
Arduino Logic gates w Transistors

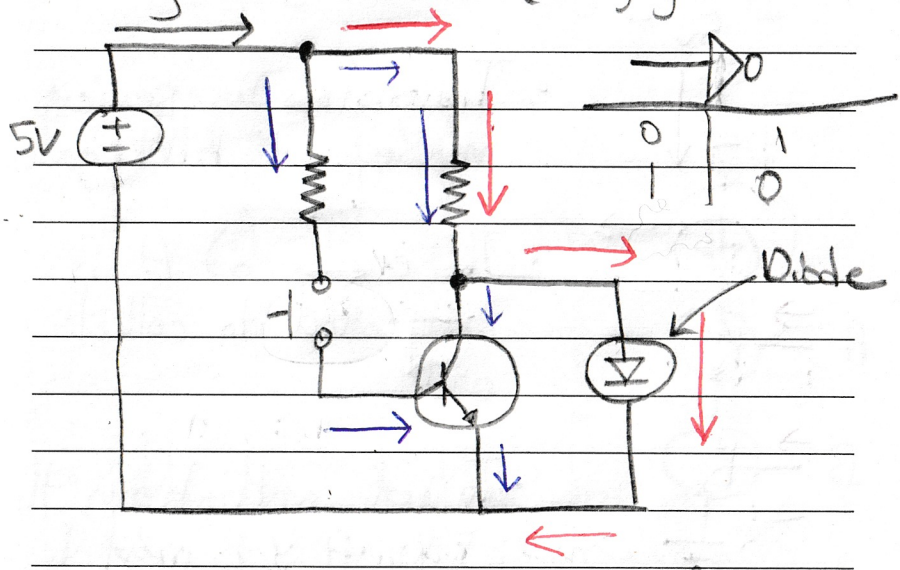
What is a transistor?



- If we have current passing through from the base and the collector, it will go through the emitter with larger current.

- If the base is recieving current it turns on the transistor, then if there is current passing through collector it goes through the emitter.

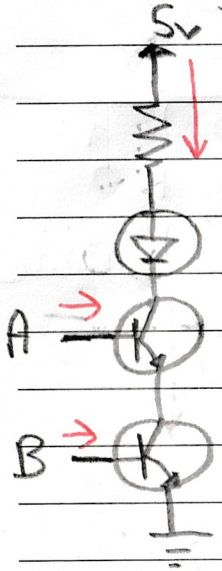
Making an inverter (Not) gate:



Case 1: The base is de activated so no current can pass through the collector. Thus the only path is red.
Red

Case 2: The base is activated so current can pass through the collector. Now very little current would pass through the LED.
Blue

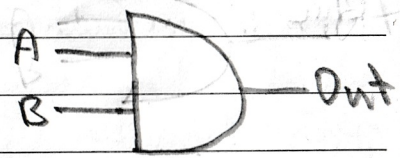
Making an AND gate:



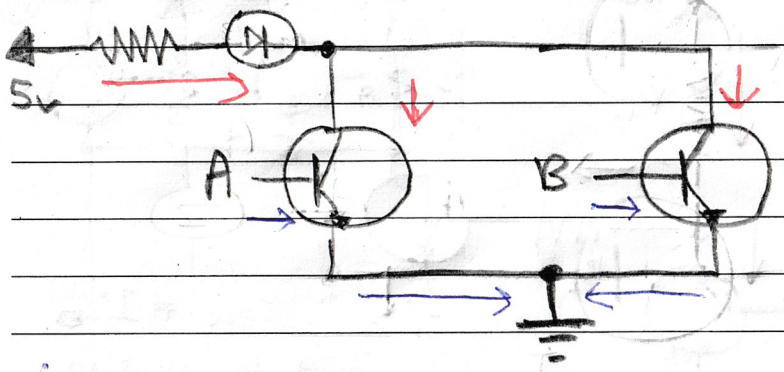
- Transistors in series make an AND gate.
- The emitter of A is connected to collector of B
- To get a High on the LED A & B must be high.

Truth table

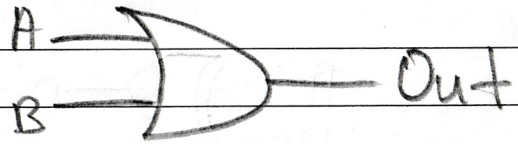
A	B	A∧B
0	0	0
0	1	0
1	0	0
1	1	1



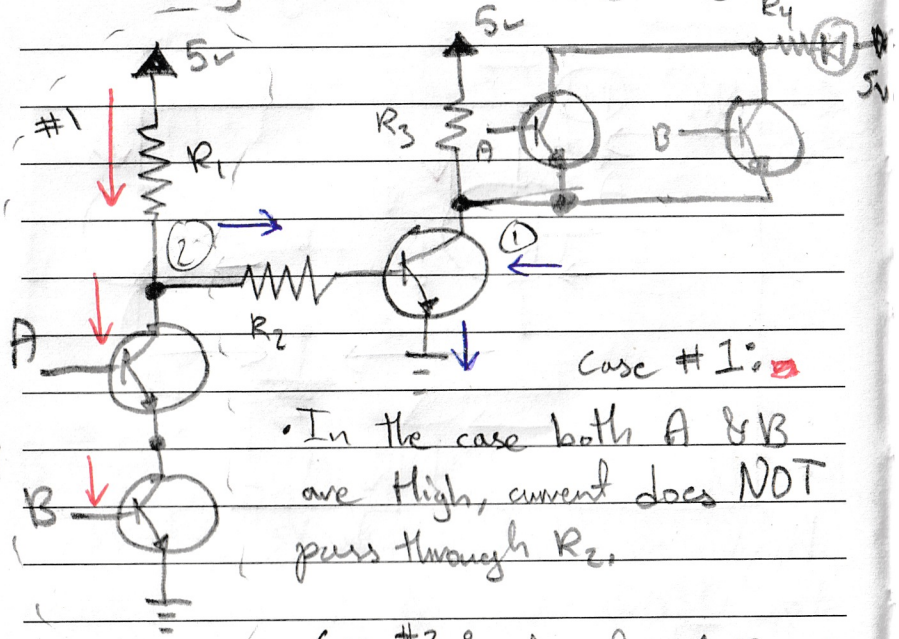
Making an OR gate:



- We are waiting for a high on base A or Base B to allow current to pass from either transistor to complete the circuit.
- Transistors in parallel make an OR Logic gate.



Making an XOR logic gate



Case #1:

- In the case both A & B are High, current does NOT pass through R_2.

- Case #2: when A and B are both off only R_2 gets power

- Case 3: if A⊕B, Point (1) is high point (2) is high, and current passes through R_2, and the circuit is closed.

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

